SECTION-I

Single Correct Answer Type **6 Q. [3 M (–1)]**

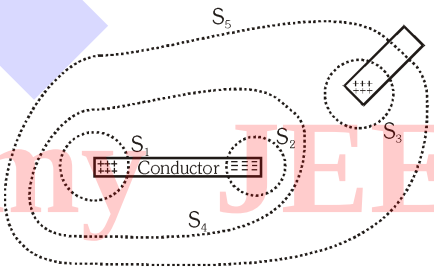
1. An electric field $\vec{E} = E_0\hat{i} + E_0\hat{j}$ exists in a space. Find the flux through a triangular loop with vertices at $\left(\frac{a}{2}, 0, 0\right)$, $\left(\frac{a}{2}, 0, a\right)$ and $\left(\frac{a}{2}, a, \frac{a}{2}\right)$

(A) $\sqrt{2}\,E_0a^2$ (B) $\frac{E_0a^2}{2}$ (C) $\frac{\sqrt{3}}{2}E_0a^2$ (D) $\frac{\sqrt{3}}{4}E_0a^2$

2. A positive point charge Q is placed at a point P, which is at a distance L from the centre 'C' of an uncharged metallic ball of radius R (L > R). There is a point G on the line PC, at a distance R/2 from C. Then, the electric potential at G is :-

(A) $\frac{Q}{4\pi\varepsilon_0 L}$ (B) $\frac{Q}{4\pi\varepsilon_0} \times \left(\frac{1}{L} - \frac{1}{R}\right)$ (C) $\frac{Q}{4\pi\varepsilon_0} \times \left(\frac{1}{L} - \frac{2}{R}\right)$ (D) None

3. A rod containing charge +Q is brought near an initially uncharged isolated conducting rod as shown. Regions with total surface charge +Q and –Q are induced in the conductor as shown in the figure. The only regions where the net charge in this configuration is non–zero are indicated by the "+" and "–" signs. Let us denote the total flux of electric field outward through closed surface S_1 as ϕ_1, through S_2 as ϕ_2, etc. Which of the following is necessarily false.



(A) $\phi_1 > 0$ (B) $\phi_2 = \phi_1$ (C) $\phi_3 = \phi_1$ (D) $\phi_4 = 0$

4. A sphere of radius R carries charge such that its volume charge density is proportional to the square of the distance from the centre. What is the ratio of the magnitude of the electric field at a distance 2R from the centre to the magnitude of the electric field at a distance of R/2 from the centre (i.e. $E_{r=2R} / E_{r=R/2}$)?

(A) 1 (B) 2 (C) 4 (D) 8

5. Consider a conductor with a spherical cavity in it. A point charge q_0 is placed at the centre of cavity and a point charge Q is placed outside conductor.

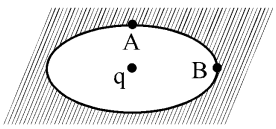
Statement-1 : Total charge induced on cavity wall is equal and opposite to the charge inside.

Statement-2 : If cavity is surrounded by a Gaussian surface, where all parts of Gaussian surface are located inside the conductor, $\oint \vec{E}.d\vec{A} = 0$; hence $q_{induced} = -q_0$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

6. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the center of the cavity. The points A & B are on the cavity surface as shown in the figure. Then :

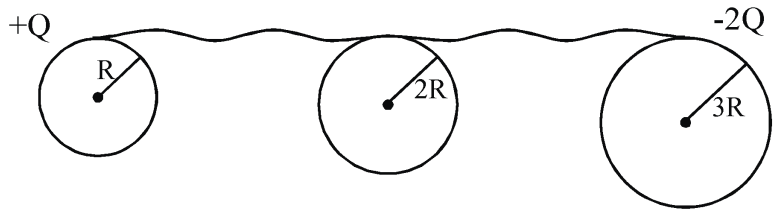
(A) electric field near A in the cavity = electric field near B in the cavity
(B) charge density at A = charge density at B
(C) potential at A = potential at B
(D) total electric field flux through the surface of the cavity is q/ε_0.



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Multiple Correct Answer Type 12 Q. [4 M (–1)]

7. Two conducting spheres of radii R and 3R carry charges Q and –2Q. Between these spheres a neutral conducting sphere of radius 2R is connected. The separation between the sphere is considerably large then :-



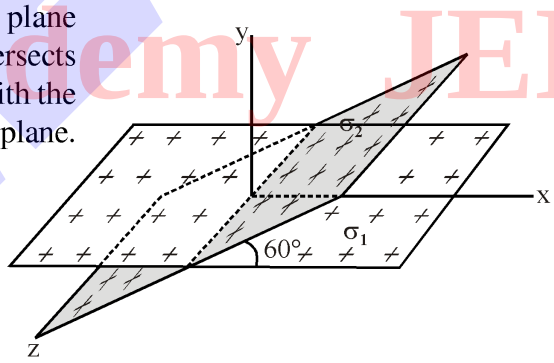
(A) The final charge on initially neutral conducting sphere is $-\frac{Q}{3}$

(B) The decrease in electric potential energy of sphere of radius R is $\frac{35KQ^2}{72R}$

(C) The decrease in electric potential energy of sphere of radius R is $\frac{37KQ^2}{72R}$

(D) The final electric potential of sphere of radius 3R will be $-\frac{KQ}{6R}$

8. An infinite plane in the xz plane carries a uniform surface charge density $\sigma_1 = 8.85$ nC/m^2. A second infinite plane carrying a uniform charge density $\sigma_2 = 17.7$ nC/m^2 intersects the xz plane at the z axis and makes an angle of 60° with the xz plane as shown in figure. The electric field in the xy plane.



(A) at x = 6m, y = 2m is $500\sqrt{3}$ N/C

(B) at x = –5m, y = 0 is $500\sqrt{3}$ N/C

(C) at x = 2m, y = 6 m is $500\sqrt{7}$ N/C

(D) at x = –1m, y = –1m is $500\sqrt{7}$ N/C

9. $\vec{E} = x\hat{i} + y\hat{j} + z\hat{k}$ exists in space then choose **INCORRECT** statement(s) :

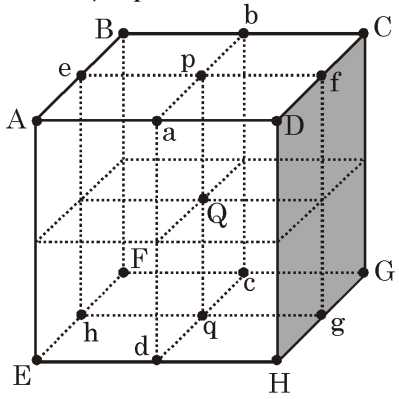
(A) This electric field is not possible

(B) Volume charge density is non uniform.

(C) If a charge particle goes from (3, 4, 5) to (5, 3, 4) then work done by electric force on charge will be positive

(D) Charge enclosed in sphere of unit radius and centered at origin is $4\pi\in_0$.

10. The following figure shows a charge Q kept at the centre of a cube. Let ϕ represent the flux of field due to the charge Q, the correct options are :-



(A) $\phi_{ABCD} = \frac{Q}{6\varepsilon_0}$

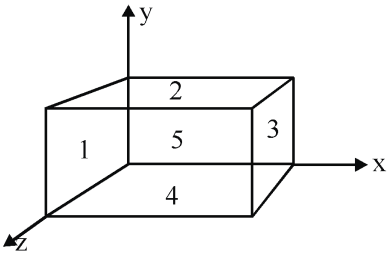
(B) $\phi_{Cbpf} = \frac{Q}{24\varepsilon_0}$

(C) $\phi_{abcd} = 0$

(D) $\phi_{abcd} = \frac{Q}{2\varepsilon_0}$

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11. In a region of space, the electric field $\vec{E} = E_0 x\hat{i} + E_0 y\hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



(A) the total electric flux through the faces 1 and 3 is E_0a^3

(B) the charge inside the cubical volume is $2\varepsilon_0E_0a^3$

(C) the total electric flux through the faces 2 and 4 is $2E_0a^3$

(D) the charge inside the cubical volume is $\varepsilon_0E_0a^3$

12. A long thin straight wire with linear charge density λ runs along axis of a thin hollow metal cylinder of radius R. The cylinder has a net linear charge density 2λ. Assume λ is positive. Mark correct options:-

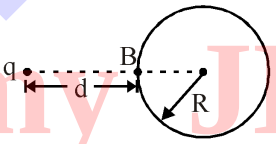
(A) $\vec{E}(r > R) = \dfrac{3\lambda}{2\pi\epsilon_0}\dfrac{\hat{r}}{r}$

(B) $\vec{E}(r < R) = \dfrac{3\lambda}{2\pi\epsilon_0}\dfrac{\hat{r}}{r}$

(C) Linear charge density on inner surface of cylinder is $-\lambda$

(D) Linear charge density on outer surface of cylinder is 3λ

13. For the situation shown in the figure below, mark out the correct statement(s)



Hollow neutral conductor

(A) Potential of the conductor is $\dfrac{q}{4\pi\varepsilon_0(d+R)}$

(B) Potential of the conductor is $\dfrac{q}{4\pi\varepsilon_0 d}$

(C) Potential of the conductor can't be determined as nature of distribution of induced charges is not known

(D) Potential at point B due to induced charges is $\dfrac{-qR}{4\pi\varepsilon_0(d+R)d}$

14. A spherical soap bubble of radius R has uniformly distributed charge over its surface with surface charge density σ then [T = surface tension of the soap solution]

(A) excess pressure inside the bubble is $\dfrac{4T}{R} - \dfrac{\sigma^2}{2\varepsilon_0}$

(B) excess pressure inside the bubble is $\dfrac{4T}{R} + \dfrac{\sigma^2}{2\varepsilon_0}$

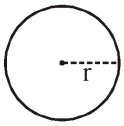
(C) excess pressure inside the bubble is $\dfrac{4T}{R}$

(D) electrostatic pressure is $\dfrac{\sigma^2}{2\varepsilon_0}$

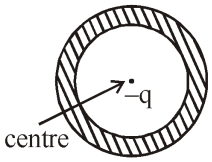
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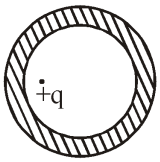
15. In which of the cases we will get uniform charge distribution of (+q) on external spherical surface. Given every object is a conductor :

(A) 

thin shell having net charge +q

(B) 

thick shell having net charge zero

(C) 

thick shell having net charge zero

(D) 

solid sphere having net charge +q

16. A conductor A is given a charge of amount +Q and then placed inside an uncharged deep metal can B, without touching it.

(A) The potential of A does not change when it is placed inside B.

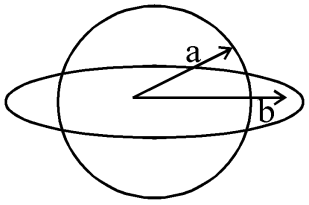
(B) if B is earthed, +Q amount of charge flows from it into the earth.

(C) if B is earthed, the potential of A is reduced.

(D) if B is earthed, the potential of A and B both becomes zero.

17. An insulating ring of charge $2\pi\lambda b$, radius b is concentric with a charged conducting solid sphere of charge Q and radius a (b >a) as shown in the figure. Then

(A) Potential of the sphere is $\frac{Q}{4\pi \in_0 a} + \frac{\lambda}{2 \in_0}$

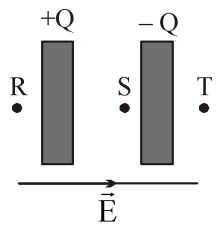


(B) If sphere is grounded then charge on the sphere will be $\frac{-a\lambda}{8\pi^2 \in_0}$

(C) Electric field inside the sphere is non-zero

(D) The sphere is equipotential body.

18. Two large thin conducting plates with small gap in between are placed in a uniform electric field E (perpendicular to the plates). Area of each plate is A and charges $+Q$ and $-Q$ are given to these plates as shown in the figure. If points R,S and T as shown in the figure are three points in space, then the



(A) field at point R is E

(B) field at point S is E

(C) field at point T is $\left(E + \frac{Q}{\varepsilon_o A}\right)$

(D) field at point S is $\left(E + \frac{Q}{A\varepsilon_o}\right)$

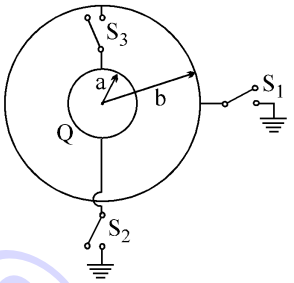
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SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

19. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' (>a). Initially switches S_1, S_2 and S_3 are open and sphere 'A' carries a charge Q. First the switch 'S_1' is closed to connect the shell B with the ground and then opened. Now the switch 'S_2' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch 'S_3' is closed to connect the spheres together. Find the heat (in joule) which is produced after closing the switch S_3. [Consider b = 4 cm, a = 2 cm and Q = 80 μC]

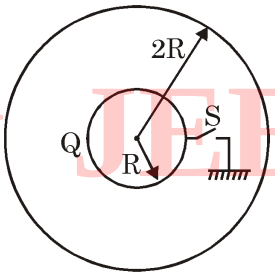


SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

20. A non-uniform electric field in x-direction is increasing uniformly from 2N/C at x = 1m to 8 N/C at x = 7m. The center of cube is at x = 2m. If the charge enclosed inside a small cube of side length 10 cm is 8.85×10^{-5n} C. The value of n will be.

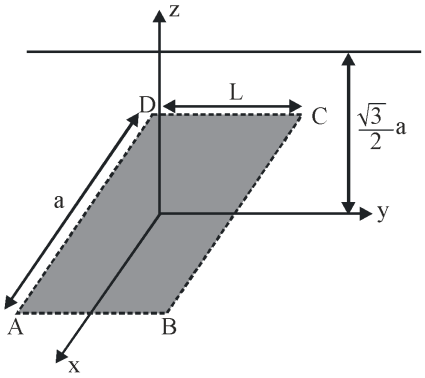
21. Two conducting spherical shell of radii R & 2R given charges Q and 2Q respectively. Inner shell is provided with a switch which can ground the inner shell, as shown. Switch is initially open and energy stored in the system is U_1. After the switch is closed, energy stored in the system is found to be U_2. Find $\frac{U_1}{U_2}$.



22. A sphere of radius R has charge density given by $\rho = \rho_0\left(1-\frac{nr}{3R}\right)$, where ρ_0 is a constant, r is distance from centre of sphere. For a spherical gaussian surface of radius R centered at the centre of sphere, the flux is zero. Find 'n'.

23. A cube is made from six thin non-conducting square faces of size a each. Each face carries a uniformly distributed charge Q. If electrostatic force acting on each face is $\frac{Q^2}{N \in_0 a^2}$, then find N.

24. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at $z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface ABCD lying in the x-y plane with its centre at the origin is $\frac{\lambda L}{n\varepsilon_0}$ (ε_0 = perimittivity of free space) then the value of n is.



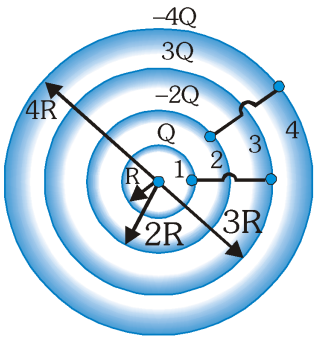
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SECTION-IV

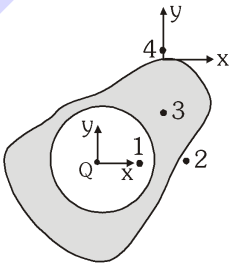
Matrix Match Type (4 × 5) **2 Q. [8 M (for each entry +2(0))]**

25. Four concentric hollow spheres of radii R, 2R, 3R and 4R have given the charges as shown. Now, the conductors 1 and 3, 2 and 4 are connected by conducting wires at the same time, then match the entries of column I with entries of column II.



Column–I	Column–II
(A) The charge an inner surface of 3rd conductor	(P) $-\frac{22Q}{5}$
(B) The charge on 4th conductor	(Q) $-\frac{Q}{8\pi\in_0 R}$
(C) The potential of conductor 1	(R) $\frac{6Q}{5}$
(D) The potential of conductor 2	(S) $\frac{-3Q}{40\pi\in_0 R}$
	(T) None of these

26. Figure shows an uncharged conducting body having a spherical cavity. Charge Q is placed at the centre of the cavity.



Column- I (Point)		Column - II (Components of electric field)	
(A)	1 (inside cavity near surface)	(P)	$E_x = 0$
(B)	2 (near but outside conductor surface)	(Q)	$E_y = 0$
(C)	3 (inside bulk of conductor)	(R)	$E_x \neq 0$
(D)	4 (near but outside conductor surface)	(S)	$E_y \neq 0$
		(T)	None of these

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GR # GAUSS LAW & CONDUCTOR — ANSWER KEY

SECTION-I

Single Correct Answer Type — **6 Q. [3 M (–1)]**

1. Ans. (B) 2. Ans. (A) 3. Ans. (B) 4. Ans. (B) 5. Ans. (A)
6. Ans. (C)

Multiple Correct Answer Type — **12 Q. [4 M (–1)]**

7. Ans. (A,B,D) 8. Ans. (A,C) 9. Ans. (A, B, C) 10. Ans. (A,B,C) 11. Ans. (A,B)
12. Ans. (A, C, D) 13. Ans. (A,D) 14. Ans. (A,D) 15. Ans. (A,C) 16. Ans. (A,C)
17. Ans. (A,D) 18. Ans. (A,D)

SECTION-II

Numerical Answer Type Question (upto second decimal place) — **1Q.[3M(0)]**

19. Ans. 180

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — **5 Q. [4 M (0)]**

20. Ans. 3 21. Ans. 5 22. Ans. 4 23. Ans. 2 24. Ans. 6

SECTION-IV

Matrix Match Type (4 × 5) — **2 Q. [8 M (for each entry +2(0)]**

25. Ans. (A) → (R) ; (B) → (P) ; (C) → (S) ; (D) → (Q)
26. Ans. (A) → (Q,R); (B) → (R,S); (C) → (P,Q); (D) → (P,S)

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